

DEE MOD I

COURSE: ELECTRICAL INSTALLATION TECHNOLOGY

COURSE CODE EET 215A

TOPIC 1: SAFETY

Workshop safety is a vital component that needs to be observed before and during any operations in the workshop.

Safety is the behaviour undertaken to prevent injury to people and damage to machines and equipment.

Reasons for observing safety

- ✓ To prevent loss of human life and or disability
- ✓ To reduce occurrence of accidents
- ✓ To protect tools and equipment in the workshop

General Safety Precautions while working in a workshop

- ✓ Always put on personal protective equipment (PPE)
- ✓ Work from marked work area
- ✓ Do not run in the workshop, always walk
- ✓ Do not work on live circuits
- ✓ Always ensure the circuit is checked for correct functioning before connecting to power supply
- ✓ All slippery floors should be cleaned and wiped
- ✓ Conceal all bare conductors
- ✓ Switch OFF any running machines and power supply before leaving the workshop
- ✓ Do not leave the machines and equipment ON even if the power supply is OFF
- ✓ Do not operate any machine unless authorised to do so
- ✓ Do not clean, oil, adjust, or repair any machine while it is still running
- ✓ Ensure work and cutting tools are clamped securely.
- ✓ Keep the floor in the workshop clean and clear from metal chips or curls and waste pieces.
- ✓ Defective guards should be replaced immediately
- ✓ Do not put on loosely hanging rings, chains, necklace etc while in the workshop
- ✓ Ensure tools and equipment are well fastened/screws tightened before using them
- ✓ Use correct tool and equipment for the right job
- ✓ When working with a ladder ensure it is properly supported
- ✓ Clear all pathways of any obstructions
- ✓ Avoid loose clothing such as necktie

PERSONAL PROTECTIVE EQUIPMENT (PPE)

PPE is defined as all equipment designed to be worn, or held, to protect against a risk to health and safety. This includes most types of protective clothing, and equipment such as eye, foot and head protection, safety harnesses, life-jackets and high-visibility clothing.

PPE includes the following:

NAME of PPE	PURPOSE
Safety Helmet	✓ Protects the head from falling objects ✓ Make sure that loosely hanging hair is tied before putting on safety helmet
Goggles/face shield	✓ Protects eyes from flying objects and excessive light
Ear muffs	✓ Protects ears from excessive sound
Dust coat/overall	✓ Protects personal clothing from dirt ✓ The protective clothing should be kept in good condition
Safety boots	✓ Protects feet from injury/falling objects ✓ Should be of a type which provides protection against slippery surfaces, and be fitted with steel toe caps and steel sole plates
Hand Gloves	✓ Protects hands from injury ✓ Industrial gloves that do not cause slippery of tools should be used

SAFETY SIGNS

These are rules and regulations of the working environment communicated by written instructions, signs and symbols. All signs in the working environment are intended to inform. They should give warning of possible dangers and must be obeyed.

They fall into four categories:

- i. **Prohibition signs:** These are must not do signs. These are circular white signs with a red border and red cross-bar. They indicate an activity which must not be done.
- ii. **Warning signs:** These give safety information. These are triangular yellow signs with a black border and symbol. They give warning of a hazard or danger.
- iii. **Mandatory instructions:** These are must do signs. These are circular blue signs with a white symbol. They give instructions which must be obeyed.
- iv. **Safe condition signs:** These give safety information. These are square or rectangular green signs with a white symbol. They give information about safety provision.

NOTE: Safety signs must be displayed in appropriate places where they can be easily seen

SAFE WORKING HABITS

1. Handling of tools and equipment

All tools and equipment should be handled with care. They should be handled individually. They should never be staked as a pile.

2. Handling of heavy loads

Always ask for assistance. Carry heavy loads with the back upright. If the load is too heavy use a crane or other mechanical assistance

3. Movement of sharp items

Sharp items should be moved pointing downwards and away from the body. The movement should be slowly. Never run when carrying sharp items. Always warn other people when carrying sharp and/or long objects

4. Working environment

The working area should be well ventilated and well lit. the floor should be free from oil or any slippery substance. Always wipe any liquid that is on the floor.

CARE AND MAINTENANCE OF TOOLS

- ✓ Tools should be stored in their racks or kits
- ✓ After use, the tools should be cleaned
- ✓ Wipe them to be free from dust and oil or grease
- ✓ Use cotton for cleaning
- ✓ Cutting tools should be sharpened from time to time
- ✓ Tools that require oiling should be oiled as planned

FIRST AID BOX

First aid is the first treatment given to an accident victim before regular medical treatment is obtained. It is important for one to familiarise themselves with basic first aid skills.

The first aid box/kit is a collection of supplies and equipment. All workshops must have a first aid kit which can be used to address cuts, burns etc.

Basic contents of the first aid box

- | | |
|--------------------|------------------------|
| ✓ Pair of scissors | ✓ Adhesive plasters |
| ✓ Pain killers | ✓ Roller bandages |
| ✓ Surgical spirits | ✓ Petroleum jelly |
| ✓ Eye drops | ✓ Sterilised dressings |
| ✓ Cotton wool | |

ELETRICAL HAZARDS/DANGERS OF ELECTRICITY

Causes of electrical hazards include:

- | | |
|--|--------------------------|
| ✓ Poor wiring and installation | ✓ Ignorance |
| ✓ Poor sizing of electrical components and cables | ✓ Overconfidence |
| ✓ Worn out insulation | ✓ Naked wires/bare wires |
| ✓ Carelessness of individuals in handling electrical equipment | ✓ Power surges |

Types of Electrical hazards

- | | |
|------------------|-------------------|
| ✓ Electric shock | ✓ Electrical Fire |
|------------------|-------------------|

ELECTRIC SHOCK

This is the passage of current through the human body when the body becomes part of a conducting path. It causes the body to tingle and /or when high currents are involved unconsciousness and/or paralysis is experienced.

Predetermining factors of the effect of electric shock on the body are:

- ✓ State of the body i.e whether wet or dry
- ✓ Amount of current in the conducting path; heavy or light current

- ✓ Path followed by the current through the body; the most dangerous path is through the chest and heart

Procedure for helping a victim of electric shock

- ✓ Use a dry insulator to remove the victim from electrical contact
- ✓ Take the victim away from the place where the shock occurred
- ✓ If the victim has stopped breathing lay him or her flat on the ground then apply first aid by kiss of life method of artificial respiration as follows:
 - i. Place a support below the neck and loosen the clothing
 - ii. Clear all air passages. Pinch the nose and breath into the victim`s mouth until the chest rises
 - iii. Release the nose and let the air rush out. Continue with this process until the victim breathes normally
 - iv. Take the victim to hospital for further medical check-up

Holger-Nielsen method of artificial respiration

- i. Lay the casualty in prone position on a flat surface
- ii. Place the casualty`s hands one over the other under his forehead
- iii. The head must be turned slightly on one side
- iv. Nose and mouth must be un obstructed
- v. Place one knee with its inner side in line with the casualty`s cheek, a few inches from the top of his head
- vi. Place the other foot with the head in line with the casualty`s elbow.
- vii. Place your hands on the casualty`s back on the lower part of the shoulders back, fingers pointing at casualty`s feet
- viii. Keeping the arms straight, rock forward gently until arms are vertical. The movement takes seconds counting one, two. This pressure causes expiration.
- ix. Then rock back counting “three” for one second and slide your hand past the casualty`s shoulder until you can grasp his upper arms near the elbows
- x. Raise and pull the arms until tension is felt for two seconds counting “four, five”. Do not raise chest from ground. This movement causes inspiration.
- xi. Your hands should remain straight count “six” for one second

FIRE - is a chemical reaction which will continue if fuel, oxygen and heat are present. To eliminate a fire one of these components must be removed.

Fires are divided into the following classes or categories:

- ✓ Class A are fires which occur due to presence of wood, paper and textile fires;
- ✓ Class B are fires which occur due to presence of flammable liquids such as paint, petrol and oil;
- ✓ Class C fires which occur due to presence of flammable gases like methane, propane etc;
- ✓ Class D are special types of fires involving burning metal like titanium, magnesium, sodium etc.
- ✓ Electrical fires are fires which occur due to poor installation, poor insulation, short circuit, overloading, faulty electrical equipment

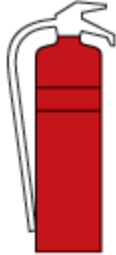
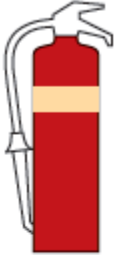
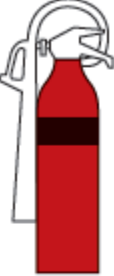
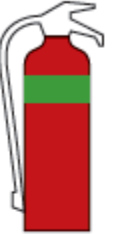
Fire detection can be done by:

- ✓ **Human being** – where an individual notices fire and manually raises the alarm at a break glass point
- ✓ **Smoke detectors** – rely sensors that automatically detect smoke and raise alarm

- ✓ **Heat detectors** – rely on sensors that automatically detect heat and raise alarm

FIRE EXTINGUISHER

Also referred to as fire suppression and it involves fighting of fire using a suitable extinguisher. Classification of fire extinguishers and their applications is as shown in the table below.

Type of fire extinguisher Type of fire	(i) Water	(ii) Foam	(iii) Carbon dioxide gas	(iv) Dry powder	(v) Vapourizing foam
	 Signal red flash on red	 Pale cream flash on red	 Black flash on red	 French blue flash on red	 Emerald green flash on red
Class A. Paper, wood and fabric	✓ Yes	✓ Yes	✗ No	✓ Yes	✓ Yes
Class B. Flammable liquids	✗ No	✓ Yes	✓ Yes	✓ Yes	✓ Yes
Class C. Flammable gases	✗ No	✗ No	✓ Yes	✓ Yes	✓ Yes
Electrical fires	✗ No	✗ No	✓ Yes	✓ Yes	✓ Yes
Motor vehicle protection	✗ No	✓ Yes	✗ No	✓ Yes	✓ Yes

Handling a victim of ELECTRICAL FIRE

Electrical fires are caused by short circuits and loose connections.

- ✓ Raise the fire alarm
- ✓ De-energize the circuits by switching OFF power if possible
- ✓ Leave the building/workshop in a briskly and orderly manner and assemble at the fire assembly point

- ✓ Put OFF the fire using recommended fire extinguishers which are carbon dioxide/foam/dry powder and other materials like dry sand and fire blanket
- ✓ Always alert fire experts/brigades
- ✓ Take roll call register to note the presence of everyone

Handling a victim of ELECTRICAL BURN

- ✓ Cool the burnt skin with cold water and spray water over any smouldering clothing
- ✓ Do not remove the clothing carelessly as it may remove the damaged skin
- ✓ Dress the wounds with burn dressing from the first aid box
- ✓ Seek medical attention

Handling a victim of CUTS AND ABRASIONS

- ✓ Clean with antiseptic and bandage to stop bleeding and infection
- ✓ If there's heavy bleeding, lay the patient down to reduce blood pressure while elevating the bleeding spot
- ✓ Expose the wound in case of blood clots apply pressure with a thick pad, apply dressing and then bandage
- ✓ Seek medical attention

Handling a victim of BRUISES

- ✓ Apply ice bergs or dump pieces of clean cloth on the area until pain stops
- ✓ Seek medical attention if pain persists

KNEC PAST PAPERS TOPIC RELATED QUESTIONS

NOV SERIES 2019

1. Explain electric shock with reference to electric safety
2. State three factors that determine the level of an electric shock
3. Explain `safe working practices` with reference to each of the following
 - i. Work areas
 - ii. Clothing
4. An electrician on site accidentally falls from a high scaffolding and is found by the technician in charge lying unconscious. Outline the first aid procedure which the technician must carry out

JULY SERIES 2019

NOV SERIES 2018

JULY SERIES 2018

NOV SERIES 2017

JULY SERIES 2017

NOV SERIES 2016

1. Name two:
 - i. types of fire extinguishers and the type of fire they are used on;
 - ii. components of a first aid box.
2. Explain three main causes of electric fires

JULY SERIES 2016

1. Explain the following with reference to safety
 - i. Electric shock
 - ii. Protective clothing
2. Describe the Holger Nelson Method carried out on an electric shock victim
3. Explain how each of the following can cause accidents
 - i. Using defective tools
 - ii. Improvising tools

NOV SERIES 2015

JULY SERIES 2015

JULY SERIES 2013

1. Explain the purpose of safety rules and regulations in workshops
2. Outline any five electrical safety rules and regulations
3. Describe the procedure to follow while attending to personnel who suffers from burns as a result of an electric shock
4. State three employee`s obligations towards achievement of safety while carrying out his duties in an electrical workshop or any other work place

JULY SERIES 2012

1. Distinguish between division 0 and 1 with respect to hazardous areas
2. With the aid of a diagram, explain how an electrical equipment installed in a division 0 area meets the IEE Regulation requirements

NOV SERIES 2011

1. Name two:
 - a. Causes of fire in workshop
 - b. Types of fire extinguishers
2. Describe the correct procedure for rescuing a victim of an electric shock